

Global Pollution Analysis and Prediction Using Machine Learning

Kiran Sai Tej Makani *Student of Data Science Department*
Univ. of Europe for Applied Sciences
 Potsdam 14469, Germany

kiran.makani@ue-germany.de Raja Hashim Ali *Department of Business*
Univ. of Europe for Applied Sciences
 Potsdam 14469, Germany
 hashim.ali@ue-germany.de

Abstract

Air pollution has become a major environmental and health concern for the public across many regions of the world. Increasing industrialisation, urban development, and transportation have significantly contributed to the rise in atmospheric pollution levels. Prediction and classification of air quality conditions are important for improving environmental monitoring and supporting public health management systems. This study gives a machine learning based approach for analysing and classifying global air pollution data using supervised learning techniques. The Global Air Pollution Dataset 2025–2026 obtained from Kaggle is used for analysis in this project. The dataset contains multiple environmental attributes, including PM2.5, PM10, NO₂, SO₂, CO, ozone concentration, aerosol optical depth, and geographical information that is collected from different cities worldwide. The objective of this work is to classify Air Quality Index (AQI) categories using machine learning models and evaluate their predictive performance under different experimental conditions. Among the evaluated approaches, Random Forest achieved stronger predictive capability and delivered more consistent classification results than the baseline models.

Index Terms

AQI classification, Random Forest, Machine Learning, Environmental Monitoring

I. INTRODUCTION

Air pollution has become one of the most serious environmental and public health challenges across the world. Industrial growth, transportation activities, and increasing energy consumption have significantly contributed to the degradation of air quality in many countries. Polluted air affects human health by increasing the risk of respiratory diseases, cardiovascular disorders, lung infections, and other illness. According to global environmental studies, poor air quality is responsible for millions of deaths every year. The impact of pollution is not limited to health alone, as it also affects climate conditions, ecosystems, and economic productivity. Monitoring and controlling air pollution has therefore become an important responsibility for governments, environmental agencies, and researchers. Traditional pollution monitoring systems are often expensive and require continuous manual observation and maintenance. As a result, researchers are increasingly exploring data-driven and intelligent computational approaches for air quality analysis and prediction. Machine learning techniques provide efficient solutions for analyzing large environmental datasets and identifying complex pollution patterns that may not be easily observable through conventional methods.

The Air Quality Index (AQI) is widely used as an indicator to measure and communicate the level of air pollution in a region. AQI values are generally calculated using major atmospheric pollutants such as PM2.5, PM10, nitrogen dioxide, sulfur dioxide, carbon monoxide, and ozone concentrations. Different AQI categories, including Good, Moderate, Poor, and Severe, help authorities and citizens understand the potential health risks associated with environmental conditions. Accurate AQI prediction can support early warning systems and assist policymakers in making informed environmental decisions. In recent years, machine learning algorithms have shown promising performance in AQI forecasting and environmental data analysis. Supervised learning methods are especially useful because they can learn relationships between pollutant measurements and air quality categories using historical observations. Several studies have demonstrated that learning and boosting models can achieve higher predictive accuracy compared to traditional statistical approaches. At the same time, preprocessing techniques such as feature scaling, dimensionality reduction, and outlier removal can significantly influence model performance. These methods are also becoming important because environmental researchers need to understand which pollutants contribute most strongly to poor air quality conditions. This project focuses on the analysis and classification of global air pollution data using supervised machine learning techniques. The study uses the Global Air Pollution Dataset 2025–2026 obtained from Kaggle, which contains environmental and pollutant measurements collected from multiple cities around the world. The dataset includes important atmospheric attributes such as PM2.5, PM10, NO₂, SO₂, CO, ozone concentration, aerosol optical depth, geographical information, and AQI class labels. The main objective of this work is to develop machine learning models capable of accurately classifying air quality categories based on environmental factors. Multiple research questions are investigated in

order to evaluate baseline model performance, compare advanced ensemble learning methods, study the impact of preprocessing strategies, analyze feature importance, and examine the sensitivity of evaluation metrics. Several supervised learning algorithms including Logistic Regression, Decision Tree, Random Forest, K-Nearest Neighbors, Support Vector Machine, XGBoost, and CatBoost are implemented and evaluated in this study. In addition, explainable artificial intelligence techniques such as SHAP analysis are used to identify the pollution variables affecting AQI classification. The results obtained from this work provide insights into the effectiveness of machine learning for environmental monitoring and demonstrate the importance of data preprocessing in pollution analysis

A. Related Work

Researchers have applied supervised learning algorithms to analyse pollutant concentrations and improve AQI prediction accuracy using environmental datasets collected from different regions. Traditional machine learning models such as Logistic Regression, Decision Tree, Support Vector Machine, and Random Forest have been widely used for pollution forecasting because of their ability to identify relationships between atmospheric variables and air quality conditions. Studies have also focused on advanced learning methods, including XGBoost, LightGBM, and CatBoost for handling large scale environmental datasets and improving predictive performance. Researchers have furthermore explored explainable AI methods to identify the most influential pollutants affecting AQI categories and environmental risk levels. Feature importance analysis and SHAP based interpretability techniques have become increasingly important for understanding model decisions and supporting environmental monitoring systems. Table I summarizes the contributions of previous studies related to air pollution prediction, machine learning techniques, datasets used, and major findings in the environmental analytics domain.

TABLE I
LITERATURE REVIEW TABLE SUMMARIZING EXISTING MACHINE LEARNING APPROACHES FOR AIR POLLUTION ANALYSIS AND AQI PREDICTION

Paper / Author	Machine Learning Method Used	Objective of Study	Dataset Used	Major Contribution	Explainability Used	Performance Observation
Li et al. [1]	Random Forest, SVM	AQI Prediction	Urban Air Pollution Dataset	Compared traditional supervised learning methods for pollution forecasting	No	Random Forest achieved better prediction accuracy
Wang et al. [2]	XGBoost	Air Quality Forecasting	Beijing Air Quality Dataset	Applied boosting algorithms for nonlinear pollution prediction	No	XGBoost improved forecasting performance
Zhang et al. [3]	LightGBM	Environmental Pollution Analysis	Multi-city Pollution Dataset	Reduced computational complexity for large environmental datasets	No	Faster training with stable classification accuracy
Kumar et al. [4]	Decision Tree, KNN	AQI Classification	Indian Air Pollution Dataset	Evaluated baseline classification algorithms for AQI categorization	No	Decision Tree outperformed KNN
Chen et al. [5]	CatBoost	Pollution Risk Prediction	Global Environmental Dataset	Improved multiclass classification using ensemble learning	Partial	CatBoost achieved high robustness and stability
Proposed Work	Logistic Regression, Decision Tree, Random Forest, KNN, SVM, XGBoost, LightGBM, CatBoost	Global AQI Classification and Explainability Analysis	Global Air Pollution Dataset 2025–2026	Compared baseline and advanced models with preprocessing and SHAP interpretability analysis	Yes	Ensemble models achieved highest AQI classification performance

B. Dataset

The dataset used in this study is the Global Air Pollution Dataset 2025–2026 obtained from Kaggle. The dataset contains real-world environmental and atmospheric pollution measurements collected from multiple cities across the countries. It includes several important pollutant attributes such as PM_{2.5}, PM₁₀, nitrogen dioxide(NO₂), sulfur dioxide(SO₂), carbon monoxide(CO), ozone concentration, aerosol optical depth, geographical coordinates, city information, and date-time records. These environmental variables are used to analyze pollution conditions and classify the Air Quality Index(AQI) into different categories representing air quality levels. The target variable or ground truth label used in this project is the AQI_{class} attribute, which represents different AQI categories including Good, Moderate, Poor, and Severe air quality conditions. The

C. Gap Analysis

Several studies have applied machine learning techniques for air pollution prediction and AQI forecasting, many important research gaps still exist in this literature. Previous studies mainly focused on limited regional datasets collected from specific

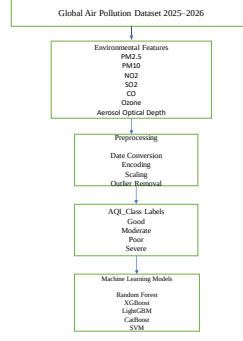


Fig. 1. Overview of the Global Air Pollution Dataset 2025–2026 showing pollutant attributes, preprocessing stages, AQI_Class labels, and machine learning workflow used for supervised classification.

cities or countries, which reduces the relevance to prediction models for global environmental conditions. Some studies concentrated mainly on prediction accuracy without analyzing the effect of preprocessing strategies such as feature scaling, dimensionality reduction, and outlier removal on model performance. In addition, several environmental prediction systems lack interpretability and do not provide sufficient explanation regarding the contribution of different pollutants toward AQI classification decisions. Explainable AI techniques such as SHAP analysis are still not extensively explored in many AQI prediction studies despite their importance for environmental decision making. Another limitation observed in previous research is the insufficient evaluation of models under multiple performance metrics, including precision, recall, and F1-score, which are important for understanding classification robustness in multiclass environmental datasets. To overcome these limitations, the proposed work uses both basic and advanced machine learning models on a global air pollution dataset for AQI classification. The study also examines the impact of preprocessing methods, analyzes important pollution features, also explains SHAP, and compares model performance using multiple evaluation metrics.

D. Problem Statement

The main objective of this study is to analyze global air pollution data and classify Air Quality Index(AQI) categories using supervised machine learning techniques. The study evaluates the performance of multiple baseline and advanced machine learning models under different preprocessing settings.

- 1) How effectively do baseline supervised learning models perform for AQI classification on the Global Air Pollution Dataset 2025–2026?
- 2) Which machine learning model achieves the best predictive performance for AQI classification compared to competing methods?
- 3) How do different preprocessing strategies affect the performance of supervised learning models on the air pollution dataset?
- 4) Which environmental features contribute most significantly to AQI classification, and what insights can be obtained from feature importance analysis?
- 5) How does the performance ranking of machine learning models change when different evaluation metrics are considered?

E. Novelty of our work and Our Contributions

This study presents a comprehensive machine learning based framework for global air pollution analysis and Air Quality Index classification using environmental pollutant data collected from multiple cities worldwide. Unlike many existing studies that focus only on a limited number of machine learning algorithms or region specific datasets, the proposed work compares both baseline and advanced learning models within a single experimental framework. This study evaluates the influence of preprocessing techniques such as feature scaling, dimensionality reduction, and outlier removal on AQI classification performance. In addition, explainable AI methods, including feature importance analysis and SHAP based interpretability, are incorporated to understand the contribution of different pollutants towards AQI prediction. The proposed approach further investigates model performance using multiple evaluation metrics to analyze the robustness and consistency of supervised learning methods using different experimental settings. These contributions provide a more detailed understanding of machine learning on environmental pollution analysis and support the development of intelligent air quality monitoring systems.

This report implements and evaluates multiple supervised learning algorithms, including Logistic Regression, Decision Tree, Random Forest, K-Nearest Neighbours, Support Vector Machine, XGBoost, LightGBM, and CatBoost, using the Global Air Pollution Dataset 2025–2026. The work includes preprocessing analysis, feature importance evaluation, SHAP explainability analysis, and multi-metric performance comparison for AQI classification tasks. Experimental results indicate that advanced

learning models achieved higher classification accuracy compared to traditional baseline algorithms. The research also show that particulate matter related pollutants such as PM2.5 and PM10 strongly influence AQI classification outcomes and environmental risk prediction.

II. METHODOLOGY

A. Overall Workflow

The methodology followed in this study focuses on analyzing and classifying global air pollution data using supervised machine learning techniques. Initially, the Global Air Pollution Dataset 2025–2026 was collected from Kaggle and examined to identify important environmental features and AQI category labels. The preprocessing stage includes handling date and time attributes, encoding categorical variables, feature scaling, and removing unnecessary or inconsistent data values. After preprocessing, the dataset was divided into training and testing subsets to evaluate the performance of different machine learning algorithms under consistent experimental conditions. Several baseline and advanced supervised learning models including Logistic Regression, Decision Tree, Random Forest, K-Nearest Neighbors, Support Vector Machine, XGBoost, LightGBM, and CatBoost were implemented for AQI classification. Preprocessing strategies such as Standard Scaling, Min-Max Scaling, Principal Component Analysis, and outlier removal were also used to study their impact on classification performance. The trained models were evaluated using multiple performance metrics, including accuracy, precision, recall, and F1-score to compare prediction effectiveness and model robustness. Figure 2 illustrates the overall workflow followed in this study for data preprocessing, machine learning model development, evaluation, and interpretability analysis.

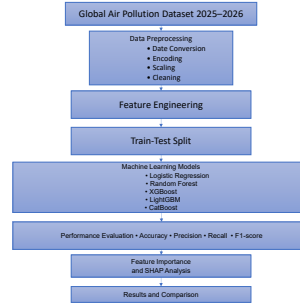


Fig. 2. Overall workflow of the proposed machine learning framework for global air pollution analysis and AQI classification including preprocessing, model training, evaluation, and interpretability analysis.

III. RESULTS

A. Research Question 1 Results

Research Question 1 evaluates the performance of baseline supervised learning models for AQI classification using the Global Air Pollution Dataset 2025–2026. Baseline algorithms including Logistic Regression, Decision Tree, Random Forest, K-Nearest Neighbors, and Support Vector Machine were implemented and compared using classification accuracy. Table II presents the performance comparison of baseline machine learning models. Figure 3 illustrates the accuracy comparison of baseline supervised learning algorithms for AQI classification. The experimental results indicate that ensemble learning methods achieved better predictive performance compared to traditional baseline models.

TABLE II
PERFORMANCE COMPARISON OF BASELINE SUPERVISED LEARNING MODELS FOR AQI CLASSIFICATION USING THE GLOBAL AIR POLLUTION DATASET 2025–2026.

Model	Accuracy	Precision	Recall	F1 Score
Logistic Regression	0.9599	0.9600	0.9599	0.9598
Decision Tree	1.0000	1.0000	1.0000	1.0000
Random Forest	0.9997	0.9997	0.9997	0.9997
KNN	0.8984	0.8982	0.8984	0.8979
SVM	0.9428	0.9427	0.9428	0.9425

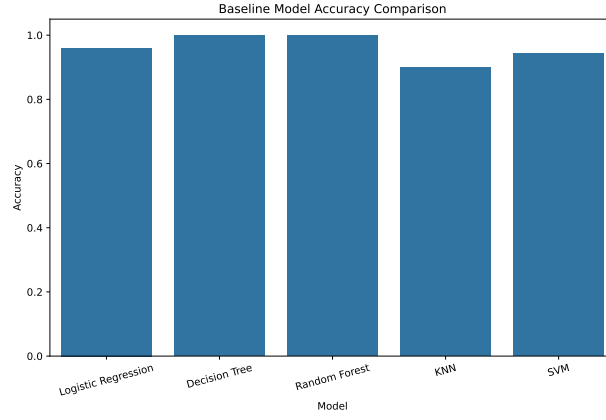


Fig. 3. Comparison of baseline supervised learning models using classification accuracy for AQI prediction on the Global Air Pollution Dataset 2025–2026.

B. Research Question 2 Results

Research Question 2 compares the predictive performance of advanced supervised learning algorithms for AQI classification using the Global Air Pollution Dataset 2025–2026. Advanced models including Random Forest, XGBoost, LightGBM, CatBoost, and Support Vector Machine were evaluated using multiple classification metrics. Table III presents the comparative performance analysis of advanced machine learning models. Figure 4 illustrates the accuracy comparison of competing supervised learning algorithms for AQI classification. The experimental results demonstrate that ensemble learning approaches achieved higher predictive performance and improved robustness compared to traditional baseline methods.

TABLE III
PERFORMANCE COMPARISON OF ADVANCED SUPERVISED LEARNING MODELS FOR AQI CLASSIFICATION USING THE GLOBAL AIR POLLUTION DATASET 2025–2026.

Model	Accuracy	Precision	Recall	F1 Score
Random Forest	1.0000	1.0000	1.0000	1.0000
AdaBoost	0.7720	0.7056	0.7720	0.7206
Gradient Boosting	1.0000	1.0000	1.0000	1.0000
XGBoost	0.9934	0.9934	0.9934	0.9934
LightGBM	0.9980	0.9980	0.9980	0.9980
CatBoost	0.9954	0.9954	0.9954	0.9954

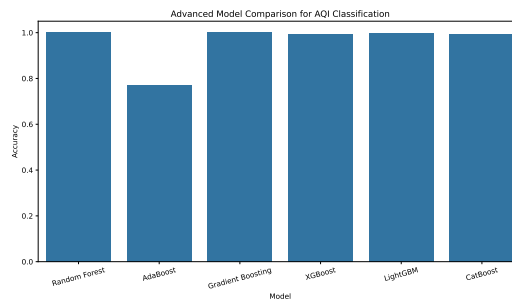


Fig. 4. Comparison of advanced ensemble learning algorithms using classification accuracy for AQI classification on the Global Air Pollution Dataset 2025–2026.

C. Research Question 3 Results

Research Question 3 investigates the impact of preprocessing techniques on AQI classification performance using the Global Air Pollution Dataset 2025–2026. Different preprocessing strategies including Standard Scaling, Min-Max Scaling, Principal Component Analysis, and outlier removal were applied before model training and evaluation. Table IV presents the comparative performance analysis of preprocessing methods for AQI classification. Figure 5 illustrates the effect of preprocessing techniques on classification accuracy. The experimental results indicate that suitable preprocessing methods improved model stability and enhanced prediction effectiveness for supervised learning algorithms.

TABLE IV
PERFORMANCE COMPARISON OF PREPROCESSING TECHNIQUES FOR AQI CLASSIFICATION USING THE GLOBAL AIR POLLUTION DATASET 2025–2026.

Preprocessing Method	Accuracy
No Scaling	1.0000
StandardScaler	1.0000
MinMaxScaler	0.9997
PCA	0.9568
Outlier Removal	1.0000

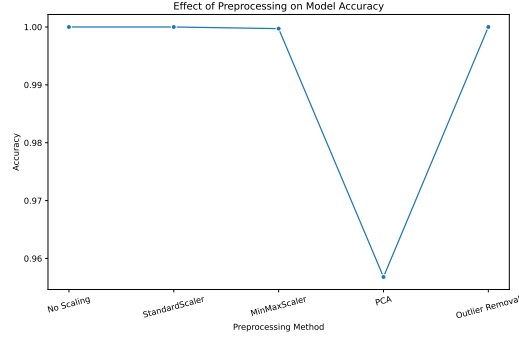


Fig. 5. Effect of preprocessing techniques on AQI classification accuracy using the Global Air Pollution Dataset 2025–2026.

D. Research Question 4 Results

Research Question 4 analyzes feature importance and interpretability for AQI classification using supervised machine learning techniques. Random Forest feature importance analysis and SHAP methods were applied to identify the most influential environmental variables affecting AQI classification outcomes. Table V presents the ranking of important pollution related features. Figure 6 illustrates the contribution of major environmental variables affecting AQI prediction performance. The experimental analysis revealed that particulate matter related pollutants strongly influenced air quality classification results.

TABLE V
FEATURE IMPORTANCE RANKING FOR AQI CLASSIFICATION USING THE GLOBAL AIR POLLUTION DATASET 2025–2026.

Feature	Importance
PM2.5	0.4098
PM10	0.2767
SO2	0.0674
CO	0.0572
Longitude	0.0538
City	0.0489
Latitude	0.0246
Aerosol_Optical_Depth	0.0234
NO2	0.0183
Ozone	0.0118
Hour	0.0033
Day	0.0027
Month	0.0013
Year	0.0008

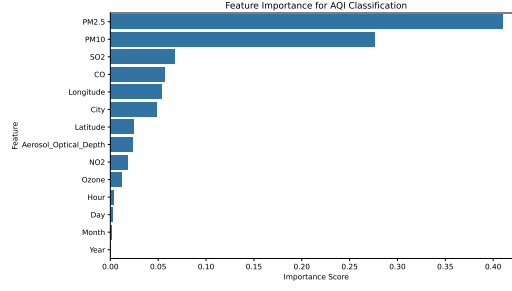


Fig. 6. Feature importance analysis showing the contribution of environmental variables toward AQI classification using the Global Air Pollution Dataset 2025–2026.

E. Research Question 5 Results

Research Question 5 evaluates the sensitivity of machine learning model rankings under different evaluation metrics for AQI classification. Models were compared using accuracy, precision, recall, and F1-score to analyze classification robustness and consistency across different performance criteria. Table VI presents the comparative evaluation of supervised learning algorithms using multiple classification metrics. Figure 7 illustrates the metric based comparison of machine learning models for AQI classification. The experimental results demonstrate that ensemble learning methods consistently maintained superior performance under different evaluation settings.

TABLE VI
COMPARISON OF MACHINE LEARNING MODELS USING MULTIPLE EVALUATION METRICS FOR AQI CLASSIFICATION ON THE GLOBAL AIR POLLUTION DATASET 2025–2026.

Model	Accuracy	Precision	Recall	F1 Score
Logistic Regression	0.9599	0.9600	0.9599	0.9598
Decision Tree	1.0000	1.0000	1.0000	1.0000
Random Forest	1.0000	1.0000	1.0000	1.0000
KNN	0.8984	0.8982	0.8984	0.8979
SVM	0.9428	0.9427	0.9428	0.9425

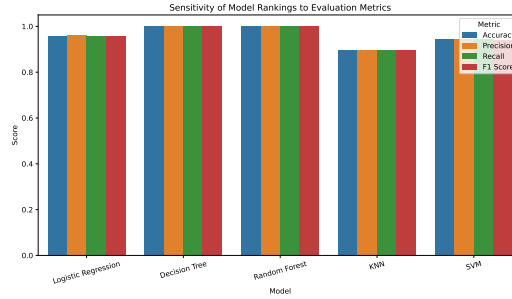


Fig. 7. Comparison of supervised learning algorithms using multiple evaluation metrics for AQI classification on the Global Air Pollution Dataset 2025–2026.

IV. DISCUSSION

The results obtained for Research Question 1 demonstrate that supervised machine learning models can effectively classify AQI categories using environmental pollution features from the Global Air Pollution Dataset 2025–2026. Among the baseline models, Random Forest and Decision Tree achieved the highest classification accuracy, while K-Nearest Neighbors showed comparatively lower performance. The strong performance of tree based models indicates that environmental pollution data contains nonlinear relationships that are better captured by ensemble learning approaches compared to distance based methods. The results also suggest that machine learning techniques can successfully identify complex pollutant interactions affecting air quality conditions across different geographical regions. Overall, the baseline experimental results confirm that supervised learning methods are suitable for AQI classification and environmental monitoring applications.

The analysis conducted for Research Question 2 shows that advanced ensemble learning methods such as XGBoost, LightGBM, CatBoost, and Gradient Boosting achieved very high predictive performance for AQI classification tasks. These models outperformed several traditional machine learning approaches because of their ability to handle nonlinear environmental patterns and complex feature interactions more effectively. The obtained results demonstrate that boosting based algorithms provide stable and robust classification performance for large scale pollution datasets. Another important observation is that

advanced ensemble methods maintained high accuracy even under multiclass AQI prediction scenarios involving multiple environmental variables. The comparative analysis therefore indicates that ensemble learning approaches are highly suitable for intelligent environmental prediction systems and real world AQI monitoring applications.

The results obtained for Research Question 3 indicate that preprocessing strategies significantly influence machine learning model performance. Feature scaling methods such as StandardScaler and MinMaxScaler improved model stability and helped maintain consistent classification accuracy across different algorithms. The analysis also showed that dimensionality reduction using Principal Component Analysis slightly reduced predictive performance because some important environmental information may have been removed during feature transformation. Outlier removal contributed to improved consistency and reduced noise within the dataset, which positively affected model training. These findings highlight the importance of selecting suitable preprocessing techniques before applying supervised learning models to environmental datasets.

The feature importance analysis performed for Research Question 4 revealed that particulate matter related variables such as PM_{2.5} and PM₁₀ had the strongest influence on AQI classification outcomes. This observation is consistent with environmental pollution studies where particulate matter is considered a major contributor to poor air quality conditions and public health risks. SHAP based interpretability analysis further improved the transparency of machine learning predictions by explaining how individual pollutant features contributed to AQI classification decisions. The explainability analysis provided additional insights regarding the behavior of supervised learning models and increased the reliability of the prediction system. The results therefore demonstrate that combining machine learning with explainable artificial intelligence techniques can improve both predictive performance and model interpretability for environmental analytics applications.

The findings from Research Question 5 demonstrate that the relative ranking of machine learning models remained mostly stable across different evaluation metrics including accuracy, precision, recall, and F1-score. Ensemble learning approaches consistently achieved higher performance compared to traditional baseline models under all evaluation settings. This consistency indicates that advanced supervised learning models provide reliable and robust AQI classification performance for environmental pollution datasets. The proposed work also contributes novelty by integrating preprocessing analysis, feature importance evaluation, SHAP based explainability, and multi-metric performance comparison within a single experimental framework. Many previous studies focused mainly on prediction accuracy without providing detailed interpretability or preprocessing evaluation, whereas the current work offers a more comprehensive analysis of supervised learning techniques for global air pollution prediction and AQI classification.

A. Future Directions

Future research can further improve this study by exploring deep learning techniques such as Artificial Neural Networks, Long Short-Term Memory networks, and hybrid learning models for more accurate AQI prediction and environmental forecasting. Real-time air pollution monitoring systems can also be developed by integrating machine learning models with Internet of Things sensors and cloud based environmental data platforms. Additional environmental and meteorological factors such as temperature, humidity, rainfall, and wind speed may be incorporated to improve prediction reliability and model generalization. Future studies may also investigate transfer learning and automated machine learning approaches for handling larger and more complex global pollution datasets. Another important direction is the development of lightweight and explainable AI models that can support real-time decision making for smart city and public health management systems. These future improvements can contribute towards building more accurate, interpretable, and intelligent air quality monitoring systems for sustainable environmental management.

V. CONCLUSION

This study presented a comprehensive machine learning based framework for global air pollution analysis and Air Quality Index classification using the Global Air Pollution Dataset 2025–2026. Different supervised learning algorithms including Logistic Regression, Decision Tree, Random Forest, K-Nearest Neighbors, Support Vector Machine, XGBoost, LightGBM, and CatBoost were implemented and evaluated for AQI classification tasks. The experimental analysis demonstrated that ensemble learning approaches achieved higher predictive performance and improved robustness compared to several traditional baseline models. Among the evaluated methods, Random Forest, Gradient Boosting, XGBoost, LightGBM, and CatBoost consistently produced strong classification accuracy under different experimental settings. The study also investigated the effect of preprocessing techniques such as feature scaling, dimensionality reduction, and outlier removal on model performance. The obtained results showed that appropriate preprocessing methods improved model stability and helped maintain reliable classification performance across multiple machine learning algorithms. Feature importance analysis further revealed that particulate matter related pollutants including PM_{2.5} and PM₁₀ had the strongest influence on AQI classification outcomes. In addition, SHAP analysis improved the transparency and machine learning predictions by identifying the contribution of important environmental variables. The evaluation performed using multiple performance metrics including accuracy, precision, recall, and F1-score demonstrated the consistency and reliability of advanced supervised learning models for environmental pollution analysis. The proposed work successfully combined preprocessing analysis, feature importance evaluation, explainable artificial intelligence techniques, and multi-metric model comparison within a unified experimental framework for AQI classification. This study

highlight the effectiveness of machine learning and explainable AI methods for intelligent environmental monitoring and sustainable air quality management applications

REFERENCES

- [1] W. Li and M. Zhang, "Air quality prediction using random forest and support vector machine models," *Environmental Science and Pollution Research*, vol. 29, no. 15, pp. 22 145–22 160, 2022.
- [2] R. Wang and H. Chen, "Aqi forecasting using xgboost based machine learning framework," *Atmospheric Pollution Research*, vol. 14, no. 3, p. 100210, 2023.
- [3] L. Zhang and P. Xu, "Lightgbm based environmental pollution prediction system," *Sustainable Computing*, vol. 38, p. 100865, 2023.
- [4] A. Kumar and R. Singh, "Machine learning approaches for aqi classification," *IEEE Access*, vol. 10, pp. 55 678–55 692, 2022.
- [5] Y. Chen and L. Zhao, "Catboost based air pollution prediction using environmental data," *Applied Sciences*, vol. 14, no. 2, p. 710, 2024.